

WELCOME ABOARD

Your MERN Journey

Six weeks from your very first line of code to a full-stack app you'll build with your team.

Duration: 6 weeks, full-time

Starting point: No coding experience needed

Where you'll finish: Shipping a full MongoDB · Express · React · Node app in TypeScript

Before you start

You're starting from zero — and that is completely fine. This plan is built for exactly that. A few things worth knowing up front:

- **Commit to GitHub every day**, starting day one. By the time it matters, it'll feel natural.
- **Ask for help early**. Being stuck for ten minutes is learning; being stuck for two hours is not. Reach out.
- **Finishing something small beats starting something huge**. A tiny app that works teaches you more than a big one that doesn't.
- **Two moments will feel hard** — `async / await` in Week 2 and your first API in Week 4. That's normal. Push through and it clicks.

Your six weeks

WEEK 1

Getting started: how software is built + your first code

WHAT YOU'LL LEARN

- How software teams actually work: the SDLC, and Agile / Scrum (sprints, backlog, user stories, stand-ups)
- Project management with **Jira** — issues, boards, and moving work from To Do to Done
- How the web works (client, server, HTTP), the command line, and **Git & GitHub**
- JavaScript basics: variables, functions, arrays, objects, loops

WHAT YOU'LL BUILD

- Set up your computer (Node.js, VS Code, Git) and create your GitHub account
- Build a Jira board with tickets for a simple practice app
- Write small JavaScript exercises and push your first repo

Hand in: A Jira board for the practice app + a GitHub repo with 5-10 JavaScript exercises.

START HERE

Atlassian Agile & Jira tutorials · MDN “JavaScript first steps” · freeCodeCamp JS · GitHub “Hello World”

WEEK 2

JavaScript in depth + TypeScript + Node.js

WHAT YOU'LL LEARN

- Modern JavaScript: arrow functions, destructuring, and `map` / `filter` / `reduce`
- Asynchronous JavaScript: promises and `async` / `await` (take your time here — it's the tricky one)
- Node.js: running JavaScript outside the browser, plus `npm` and packages
- Intro to TypeScript: what types give you, basic types, and interfaces

WHAT YOU'LL BUILD

- Rewrite earlier exercises using array methods and `async` / `await`
- Build a small Node script that fetches data from a public API
- Convert one file to TypeScript and fix the errors it surfaces

Hand in: A small Node + TypeScript script pushed to GitHub.

START HERE

MDN “Asynchronous JavaScript” · Node.js “Introduction” · The TypeScript Handbook (The Basics)

WEEK 3

Databases: SQL vs. NoSQL, and MongoDB

WHAT YOU'LL LEARN

- What a database is and why every app needs one
- **SQL vs. NoSQL** — tables and rows vs. documents and collections, and when to use each
- MongoDB fundamentals: documents, collections, and CRUD
- Mongoose with TypeScript, and connecting your Node app to MongoDB

WHAT YOU'LL BUILD

- Set up a free MongoDB Atlas database
- Practice creating, reading, updating, and deleting data
- Write a Node + TypeScript + Mongoose script that does all four

Hand in: Your own explanation of SQL vs. NoSQL in a README, plus a working CRUD script.

START HERE

MongoDB University “MongoDB Basics” · Official MongoDB & Mongoose docs

WEEK 4

MILESTONE

CRUD, REST APIs & your first backend

WHAT YOU'LL LEARN

- REST APIs: HTTP methods (GET, POST, PUT, DELETE) and status codes
- Express with TypeScript: routes, middleware, and JSON
- Connecting your API to MongoDB, plus project structure and error handling
- Testing your endpoints with Postman or Thunder Client

WHAT YOU'LL BUILD

- Build a full CRUD REST API (for example, a task list or book library)
- Test every endpoint and write a clear README
- Keep your secrets out of Git with a `.gitignore`

Hand in: Your first real backend — a Node + Express + TypeScript + MongoDB CRUD app on GitHub.

START HERE

Official Express docs · MDN “HTTP methods” and “status codes”

WEEK 5

React with TypeScript (the frontend)

WHAT YOU'LL LEARN

- What React is, and building UIs from components
- Props, state, and the core hooks: `useState` and `useEffect`
- React with TypeScript: typing your props, state, and events
- Fetching data from a backend, handling forms, and basic routing

WHAT YOU'LL BUILD

- Build small UIs: a counter, a list, a form
- Connect a React app to **your own Week 4 API** — list, add, and delete items

Hand in: A React + TypeScript frontend that does CRUD against your own API.

START HERE

react.dev “Learn React” · React + TypeScript Cheatsheet · Vite docs

WEEK 6

MILESTONE

Final project — build together

WHAT YOU'LL LEARN

- Working as a team with Git: branches, pull requests, and code review
- Planning your app in Jira and running it in short sprints (yes — the Week 1 skills come back!)
- Bringing backend and frontend together into one working product

WHAT YOU'LL BUILD

- As a team: pick an app, scope the features, and split the work into Jira tickets
- Build, integrate, and test your full MERN app together
- Present the part you built on demo day

Hand in: A full-stack MERN app in a shared repo — with your commits, your feature, and a group demo. Stretch goal: deploy it to the internet.

START HERE

Your mentors will guide the kickoff — this week is about building, not new tutorials.

Your deliverables at a glance

- ☐ **Week 1:** A Jira board for the practice app + a GitHub repo with 5–10 JavaScript exercises.
- ☐ **Week 2:** A small Node + TypeScript script pushed to GitHub.
- ☐ **Week 3:** Your own explanation of SQL vs. NoSQL in a README, plus a working CRUD script.
- ☐ **Week 4:** Your first real backend — a Node + Express + TypeScript + MongoDB CRUD app on GitHub.
- ☐ **Week 5:** A React + TypeScript frontend that does CRUD against your own API.
- ☐ **Week 6:** A full-stack MERN app in a shared repo — with your commits, your feature, and a group demo. Stretch goal: deploy it to the internet.

Every deliverable goes on GitHub — that's your growing portfolio. By Week 6 you'll have gone from your first variable to a full-stack app built with a team. That's a real developer's toolkit. Enjoy the ride.